TEST CODE **02120032**

FORM TP 2004277

MAY/JUNE 2004

CARIBBEAN EXAMINATIONS COUNCIL

ADVANCED PROFICIENCY EXAMINATION

ENVIRONMENTAL SCIENCE

UNIT 1 : ECOLOGY, PEOPLE AND NATURAL RESOURCE USE

PAPER 03/B

2 hours

01 JUNE 2004 (p.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. **DO NOT** open this examination paper until instructed to do so.
2. This paper consists of **THREE** questions. Answer **ALL** questions.
3. Write your answers in the answer booklet provided.
4. Graph paper is provided.
5. The use of a non-programmable calculator is allowed.

Answer ALL the questions.

Read the paragraph below and answer the questions that follow.

Mangrove wetlands are found in saline or brackish water and they form the basis of a complex marine food chain. They constitute a breeding habitat for many species and offer some protection for maturing offspring. Mangroves filter and assimilate pollutants in runoff, improving water quality. They stabilise bottom sediments and protect shorelines from erosion and other effects of strong winds and waves. The pressure on mangroves is due to a desire to use the coast for other purposes (human settlements, tourism) as well as the many uses of mangroves, including the production of charcoal and construction material. Infrastructure developments and buildings threaten these wetlands, both directly and indirectly.

A developer has proposed to construct a hotel in a coastal area and has identified two coastal mangrove wetlands that are suitable. The Environmental Protection Agency has requested that an inventory and a species analysis be conducted as a part of an Environmental Impact Assessment.

1. (a) Describe ONE suitable method for sampling the plant and animal populations in the two sites. **[12 marks]**

(b) Justify your choice of sampling method. **[12 marks]**

(c) Explain THREE limitations of your choice of sampling method. **[6 marks]**

Total 30 marks

2. The results of sampling the vegetation in the two sites, A and B, are shown in Table 1.

TABLE 1: RESULTS OF SAMPLING

	Number of Individuals	
Species of Vegetation	**Site A**	**Site B**
Avicennia germinans	60	5
Laguncularia racemosa	4	58
Conocarpus erectus	15	15
Rhizophora mangle	4	17

(a) Present the data in Table 1 in a suitable graphical manner. **[13 marks]**

A graph sheet is provided if needed.

(b) (i) Using an appropriate equation, calculate the species diversity for EACH of the two sites.

(ii) Which of the two sites has the greater species diversity? **[17 marks]**

Total 30 marks

3. (a) Explain the difference between 'species abundance' and 'species diversity'.
[4 marks]

(b) Immediately after the study was carried out, a hurricane devastated both sites. Based on the results of the calculations in Question 2, state which of the two sites is expected to recover faster from the disturbance. Explain your response.
[14 marks]

(c) Outline TWO measures that can be implemented to conserve mangrove ecosystems in the Caribbean. In your response, indicate why the measures you choose are appropriate.
[12 marks]

Total 30 marks

END OF TEST

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Answer sheet for Question 2 (a). **Candidate number** ....

ATTACH THIS ANSWER SHEET TO YOUR ANSWER BOOKLET